

University of Leeds
School of Electronic and Electrical Engineering

ELEC5870M Interim Report

Fabrication of Enhancement Mode HEMT Devices via Gate Recession

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Abstract

Abstract goes here

Acknowledgements

Abbreviations

2DEG	Two-dimensional electron gas
HEMT	High-electron-mobility transistor

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CHAPTER 1

Introduction

Modern communication, sensing, and microwave systems impose increasingly demanding requirements on the semiconductor devices that form their front-end signal paths. Higher operating frequencies, tighter noise constraints, and strict linearity targets continue to push transistor technologies beyond the capabilities of many silicon-based platforms. [1] Although silicon processes have made notable progress in recent years, they often fall short in providing the necessary electron mobility, noise performance, or high-frequency gain required for performance-critical microwave and millimetre-wave applications. (e.g. silicon's $1200 \text{ cm}^2/\text{Vs}$ compared to GaAs's $6500 \text{ cm}^2/\text{Vs}$ in transistors for MMICs) [1]

III–V High–Electron–Mobility Transistors (HEMTs) play a central role in addressing these challenges. Their heterostructure–based design creates a highly conductive channel at the interface between two semiconductor layers with different bandgaps. At this interface a two–dimensional electron gas (2DEG) forms, supporting exceptionally high carrier mobility. The resulting strong transconductance, stable high–frequency behaviour and low–noise performance have firmly established HEMTs as core devices throughout microwave and millimetre–wave circuit design. [2]

Research activity within the field has increasingly shifted toward gallium nitride (GaN) HEMTs, driven by the material properties enabled by its wide bandgap. GaN supports high breakdown voltages, strong power–handling capability and robust operation under large electric fields, making it the preferred choice for many high–power and high–linearity applications [3]. Despite this growing dominance, gallium arsenide (GaAs) HEMTs remain one of the most mature and well characterised III–V transistor platforms. Their stability, extensive documentation and long industrial history continue to make them an invaluable system for studying the relationships between heterostructure design, device architecture and electrical behaviour. [4, 5]

For the purposes of this project, GaAs presents clear practical and technical advantages. The Leeds Nanotechnology Cleanroom maintains established, reliable processes for GaAs/AlGaAs heterostructures, and suitable wafers are readily available with reproducible epitaxial quality [6].

Standard GaAs/AlGaAs HEMTs are inherently depletion-mode devices (d-mode); they naturally conduct current at zero gate bias and require a negative voltage to switch off. [2] This 'normally-on' behaviour is often undesirable as it necessitates complex negative polarity power supplies [7] and complicates the design of direct-coupled logic circuits. [8] Consequently, there is significant interest in developing enhancement-mode devices (e-mode) that are 'normally-off' and offer reduced power consumption.

One common approach to achieving e-mode operation is to recess the gate into the AlGaAs donor layer, thereby reducing the amount of charge carriers available to form the channel and shifting the threshold voltage in the positive direction. [9] However, achieving e-mode operation through gate recessing is inherently non-trivial, as it requires precise and repeatable removal of only a few nanometres of material, with small variations in etch depth leading to significant changes in threshold voltage and device behaviour. [10]

The primary objective of this project is to investigate the relationship between gate recess depth and threshold voltage in GaAs HEMTs, with a focus on understanding the practical limitations and repeatability of the fabrication process. This will be achieved through the development of a gate recession process in which the gate contact is recessed into the AlGaAs donor layer to shift the threshold voltage into the positive region.

To provide a rigorous comparison between d-mode and e-mode devices, this interim report first focuses on the fabrication and characterisation of a series of non-recessed d-mode devices, achieved in first semester. (Chapters 3 and 4). These standard d-mode devices serve as a control to validate the HEMT fabrication process and establish reference parameters throughout this project. Finally, Chapter 5 outlines the theory and proposed process strategy required to implement the precise donor layer etch needed to achieve e-mode operation in the second phase of this work.

CHAPTER 2

Technical Background

2.1 Overview of GaAs/AlGaAs HEMTs

In the GaAs/AlGaAs HEMTs used in this project, the interface between the GaAs channel layer and the overlying AlGaAs barrier produces a discontinuity in the conduction band due to the differing bandgaps of the two materials. This offset creates a potential well on the GaAs side of the junction, as illustrated in Figure 2.1, which is responsible for carrier confinement and the formation of a two-dimensional electron gas (2DEG) at the heterointerface. Electrons supplied by silicon donors in the modulation-doped AlGaAs layer transfer into this well and accumulate to form the 2DEG, which serves as the conducting channel of the device. The formation of this spatially confined, high-mobility channel is the defining feature of the HEMT structure and underpins its superior high-frequency and low-noise performance compared to bulk-channel transistors [11].

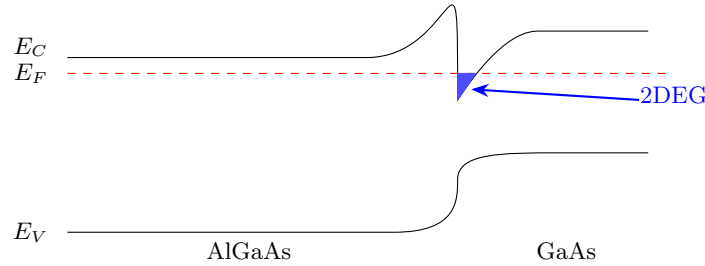


Figure 2.1: Band Structure Diagram of the AlGaAs/GaAs interface demonstrating the resulting Two-Dimensional Electron Gas (2DEG) formed in a Potential Well.

2.2 The GaAs/AlGaAs Heterostructure

The HEMTs investigated in this project are based on a conventional GaAs/AlGaAs heterostructure in which a sequence of epitaxial layers is engineered to create and control a 2DEG. A schematic cross-section of the structure is shown in Figure 2.2, highlighting the cap, donor, spacer, and channel layers whose thicknesses and compositions govern the electrostatic control and transport behaviour of the device.

At the surface of the structure lies a 10 nm thick, highly doped GaAs cap layer. This cap provides a chemically stable surface for forming the Schottky gate contact whilst also protecting the underlying AlGaAs from oxidation. Its high conductivity reduces series resistance in the access regions beneath the source and drain contacts, improving current injection into the channel.

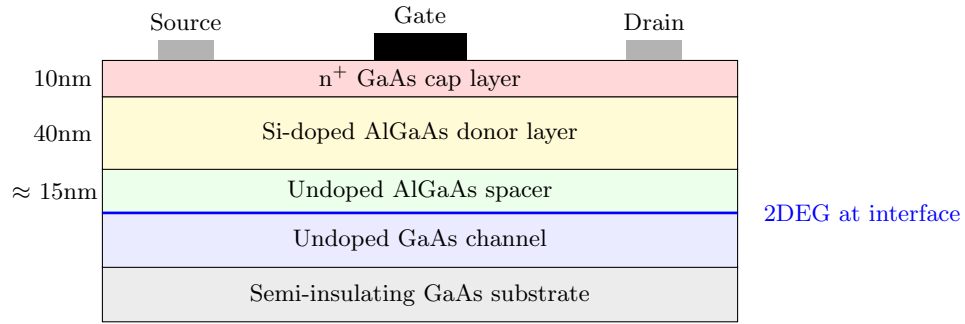


Figure 2.2: Cross-section of the GaAs/AlGaAs HEMT used in this project (not to scale).

Beneath the cap layer is a 40 nm thick silicon-doped AlGaAs donor layer. Silicon donors act as electron suppliers, with carriers transferring into the GaAs channel as a result of the conduction band discontinuity at the heterointerface. This approach, known as modulation doping, spatially separates the ionised dopants from the conducting channel. Both the donor concentration and the layer thickness strongly influence the resulting 2DEG sheet density and therefore the threshold voltage and maximum drain current of the device.

A 15 nm thick undoped AlGaAs spacer layer separates the donor layer from the GaAs channel. This spacer plays a critical role in device performance by physically isolating the 2DEG from ionised dopants, thereby suppressing Coulomb scattering [12]. This separation enables exceptionally high electron mobility, particularly at cryogenic temperatures where phonon scattering is reduced and impurity scattering would otherwise dominate transport behaviour [13]. Increasing the spacer thickness improves mobility but weakens electrostatic coupling between the gate and the channel; this trade-off is a central consideration in heterostructure design.

In this project, the donor layer thickness and the resulting effective separation between the gate and the channel represent the primary controllable parameters through which the threshold voltage may be engineered. The following section therefore focuses specifically on how gate bias modulates channel charge and determines the operational mode of the device.

2.3 Gate Control of Channel Charge and Threshold Voltage

The electrical behaviour of the GaAs/AlGaAs HEMT is governed by the field-effect interaction between the Schottky gate and the 2DEG. When a metal gate is deposited onto the semiconductor surface, a depletion region forms beneath it; the depth of this region is modulated by the applied gate voltage (V_G). This modulation controls the sheet carrier density (n_s) in the channel and, consequently, the drain current [14].

For conventional depletion-mode devices with relatively thick barrier layers, this behaviour can be approximated using a classical parallel-plate charge-control model. However, as the gate-to-channel separation is reduced towards the nanometre scale required for enhancement-mode operation, additional physical effects must be considered. In particular, the finite spatial extent of the 2DEG within the quantum well modifies the effective electrostatic coupling between the gate and the channel, motivating the refined charge-control treatment introduced by Lee *et al.* [15].

2.3.1 Finite Charge Distribution and Effective Gate Separation

In an idealised classical model, the 2DEG is assumed to reside at an infinitesimally thin sheet located precisely at the GaAs/AlGaAs interface. In reality, electrons occupy a quantised triangular potential well on the GaAs side of the heterointerface, resulting in a finite charge distribution whose centroid is displaced from the interface.

Lee *et al.* account for this effect by introducing an effective gate-to-channel separation of $d + \Delta d$, where Δd represents a quantum-mechanical correction associated with the finite extent of the electron wavefunction. For GaAs/AlGaAs HEMTs, this correction is of order 80 Å (8 nm) [15]. Incorporating this correction modifies the effective gate capacitance and may be expressed through the transconductance parameter β as

$$\beta = \frac{\epsilon \mu W}{(d + \Delta d)L} \quad (2.1)$$

In the context of this project, Δd does not represent a fundamental performance limit; rather, it acts as an effective offset that becomes increasingly relevant as the physical barrier thickness d is reduced through gate recessing. For the baseline depletion-mode devices fabricated in Semester 1 ($d \approx 50$ nm), the quantum correction constitutes a relatively small fraction of the total separation. However, as the donor layer is thinned towards the ~ 25 nm regime required for enhancement-mode operation, Δd becomes a non-negligible contribution to the overall electrostatic spacing. Its primary significance at this stage lies in informing the practical target recess depth required to achieve a positive threshold voltage, rather than in limiting transconductance scaling.

2.3.2 Threshold Voltage and Mode Determination

The distinction between depletion-mode (D-mode) and enhancement-mode (E-mode) operation is determined by the balance between the built-in potential of the Schottky gate and the thickness of the doped AlGaAs donor layer. Within the analytical framework presented by Lee *et al.* [15], the threshold voltage (V_{off}) may be expressed as

$$V_{off} = \phi_b - \Delta E_c - \frac{q N_d d_d^2}{2\epsilon} \quad (2.2)$$

where ϕ_b is the Schottky barrier height, ΔE_c is the conduction band offset, N_d is the donor concentration, and d_d is the donor layer thickness.

The quadratic dependence of V_{off} on d_d highlights the extreme sensitivity of the threshold voltage to small variations in etched depth. This relationship directly underpins the two operational modes explored in this project:

Depletion Mode (Semester 1): The baseline devices fabricated in Semester 1 employed the full epitaxial structure, comprising the 10 nm GaAs cap and the 40 nm AlGaAs donor layer. With a total gate-to-channel separation of approximately 50 nm, the depletion region does not intersect the 2DEG at zero gate bias. As a result, the channel remains populated ($V_{off} < 0$), and the device is normally on, requiring a negative gate voltage to achieve pinch-off [5].

Enhancement Mode (Semester 2 Objective): To achieve normally-off behaviour

($V_{off} > 0$), the donor layer must be thinned until the zero-bias depletion region fully depletes the 2DEG. For the donor concentration of $1 \times 10^{18} \text{ cm}^{-3}$ used in this project, Lee *et al.* identify a critical donor thickness of approximately 25 nm (250 Å) [15]. Recessing the gate to this depth shifts the threshold voltage into the positive regime, such that a positive gate bias is required to re-populate the channel and initiate conduction [9].

This strong dependence of threshold voltage on etched depth implies that fabrication non-idealities affecting the effective recess depth, such as surface oxidation and etch incubation effects, will directly translate into device-to-device variability. These considerations motivate the detailed examination of wet-etch behaviour and oxide control strategies presented in the following fabrication chapter and form a central focus of the enhancement-mode process development outlined in Chapter 5.

CHAPTER 3

Fabrication of Depletion-Mode HEMT Devices

3.1 Overview of the Fabrication Workflow

The depletion-mode HEMTs characterised in this report were fabricated using the standard III-V device process established within the Leeds Nanotechnology Cleanroom. This workflow follows the routine sequence used for GaAs/AlGaAs transistors in this facility: it comprises of 3 main stages: mesa definition, ohmic contact formation, and gate formation. This well-established procedure was followed without modification and carried out over four full-day laboratory sessions. The devices produced through this workflow provide a necessary baseline; they allow for a reliable comparison against which the behaviour of e-mode devices with different structures and dimensions can be evaluated later in this project.

Day 1 - Mesa Formation	Cleaning	Lithography	Wet Etching	Cleaning		
Day 2 - Ohmic Contacts	Cleaning	Lithography	Metal Deposition	Metal Lift-off	Cleaning	
Day 3 - Gate Formation	Annealing	Cleaning	Lithography	Metal Deposition	Lift-off	Cleaning
Day 4 - Electrical Testing	Microscope Inspection	Defect Removal	Electrical Characterisation			

Figure 3.1: Timeline overview of the four-day cleanroom workflow used to fabricate and characterise baseline depletion-mode GaAs/AlGaAs HEMTs.

3.2 Mesa Definition and Isolation Etch

The fabrication process began with the electrical isolation of individual device regions through mesa etching. Prior to lithography, the GaAs wafer was cleaned sequentially in acetone, isopropanol (IPA), and deionised water (DI) to remove organic residues and ensure sufficient photoresist adhesion. To further enhance the bonding interface between the substrate and the resist, a Surpass primer was spun onto the wafer. This priming step is critical for wet-etching applications: it modifies the surface tension to prevent the etchant from penetrating under the resist edges (an effect known as undercutting) which would otherwise lead to the delamination of the mask. A positive photoresist (S1803) was then spun onto the sample at 3000 rpm, soft-baked on a hot-plate at 115 °C, and subsequently patterned with exposure from the Heidelberg MLA 150 Maskless Aligner (MLA).

Following exposure, the sample was immersed in a caustic Microposit developer solution for 45 seconds, followed by a thorough 2 stage DI water rinse and a nitrogen blow-dry to reveal the mesa patterns. The mesa pattern was then transferred into the wafer using a sulphuric acid,

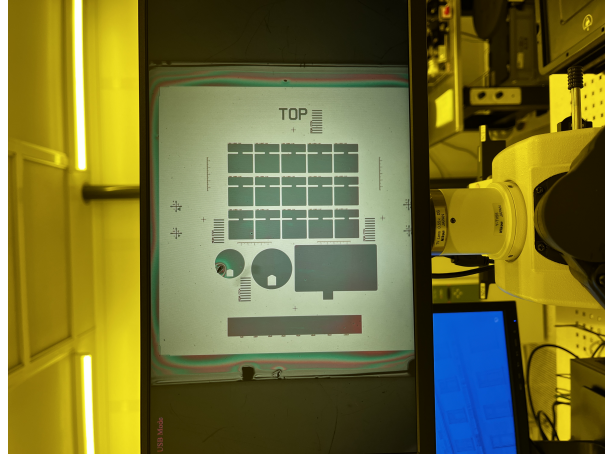


Figure 3.2: Mesa lithography mask used for defining isolated device regions (Post-Development).

hydrogen peroxide, and water wet-etch solution ($H_2SO_4 : H_2O_2 : H_2O$ in a 1:8:160 ratio). This etchant is highly suited to GaAs/AlGaAs mesa formation: it produces a controllable, moderately fast isotropic etch that removes the active layers uniformly. The rate is sufficiently slow to allow for an accurate timed etch of a shallow ≈ 200 nm recess, yet it remains fast enough to be practical for routine device isolation.

To calibrate the process, an initial 30 s pilot etch was performed; subsequent measurement using an Alpha-Step profilometer revealed an etch depth of 114 nm, corresponding to an initial etch rate of 3.8 nm s^{-1} . To reach the target depth of 200 nm, a further 86 nm of material required removal. Assuming a constant etch rate, a second etch duration of 22 s was calculated and performed; however, the final measured depth reached only 177 nm. This indicates that the second etch only removed an additional 63 nm at a reduced effective rate of approximately 2.9 nm s^{-1} . While a total depth of 177 nm is sufficient for device isolation (as it successfully penetrates the active GaAs/AlGaAs heterostructure layers), the discrepancy in rates highlights the intrinsic sensitivity of the wet-etch chemistry.

A theory for the observed reduction in the mean etch rate during the second step is attributed to the formation of a native oxide layer on the GaAs surface during the interval in which the sample was removed for profilometry. This oxide layer creates an "incubation period" at the start of the second etch: sample etching is delayed while the etchant penetrates the oxide before reaching the underlying semiconductor. Because the second etch (22 s) was shorter than the first (30 s), this non-productive period represented a larger percentage of the total etch time; consequently, the average rate for that specific step was significantly lowered.

[PROVIDE LITERATURE BACKUP TO THIS THEORY SUGGEST MITIGATION]

3.3 Ohmic Contact Formation

The formation of low-resistance Ohmic contacts is a critical prerequisite for accessing the two-dimensional electron gas (2DEG) within the heterostructure. When metal is deposited directly onto the GaAs cap, the interface naturally forms a Schottky barrier due to Fermi-level pinning at the surface states: this creates a rectifying junction that restricts carrier injection. To overcome

this, a specific metallisation scheme and thermal processing step are required to facilitate field-emission (tunnelling) conduction. The objective is to create a heavily doped n^+ region beneath the contact pads that extends down towards the channel; this high doping density significantly narrows the depletion width of the barrier, allowing electrons to tunnel freely between the metal and the semiconductor with linear $I - V$ characteristics.

To define these contacts, a dual-layer photoresist stack comprising Lift-Off-Resist (LOR) beneath S1803 was employed. A single photoresist layer typically develops near-vertical sidewalls: this causes the evaporated metal to form a continuous film across both the wafer surface and the photoresist edges, making lift-off unreliable and often leaving metallic "fences" around the contact perimeter. The dual-layer stack mitigates this by exploiting the higher developer solubility of the LOR base layer. During development, the LOR recedes laterally more than the S1803 above it, producing a distinct undercut profile. This physical overhang breaks the continuity of the deposited metal film, ensuring that the metal on top of the photoresist is electrically and mechanically separated from the device contacts. Consequently, solvent-based lift-off yields sharply defined openings free of edge defects.

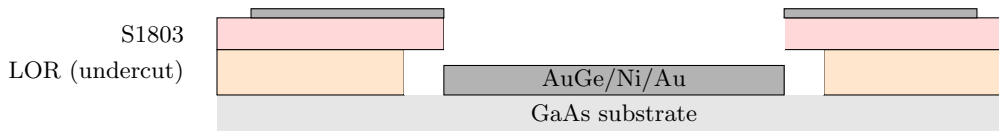


Figure 3.3: Dual-layer LOR/S1803 resist stack after development showing the wider LOR opening that creates an undercut for clean metal lift-off.

Following lithography, the metallisation was performed by the thermal evaporation of a AuGeNi eutectic pellet. The composition of this alloy is functionally specific: Germanium (Ge) serves as the amphoteric dopant which, upon substituting into Gallium sites, acts as a donor; Nickel (Ni) acts as a wetting agent to prevent the Gold-Germanium from "balling up" during the melt and assists in the diffusion process; and Gold (Au) provides the necessary low-resistance conductivity.

To activate the contacts, the sample underwent Rapid Thermal Annealing (RTA). It is during this thermal cycle that the transition from rectifying to Ohmic behaviour occurs. The heat drives the Germanium atoms to diffuse through the AlGaAs layers and into the GaAs lattice, locally creating the requisite n^+ doping profile. This effectively "thins" the potential barrier at the interface, establishing a low-resistance tunnelling path that connects the surface metallisation electrically to the buried 2DEG.

3.4 Gate Formation and Final Inspection

With the Ohmic contacts established and annealed, a third photolithography step was performed to define the Schottky gate. As with the previous stage, a dual-layer resist stack was utilised to ensure a clean lift-off. The gate electrode was aligned centrally between the source and drain contacts; its placement is critical, as it provides the electrostatic control required to modulate

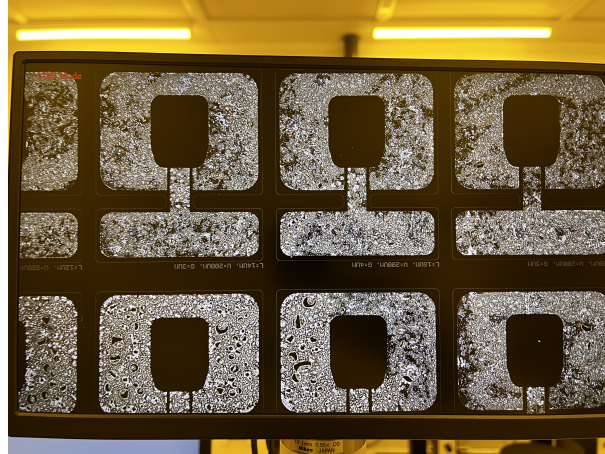


Figure 3.4: Ohmic contacts after lift-off.

the carrier density within the channel.

Following development, a Ti/Au stack was evaporated to form the gate electrode. In this configuration, the Titanium (Ti) layer serves as a vital adhesion layer to the semiconductor surface, while the thick Gold (Au) layer provides a stable Schottky barrier and minimises gate resistance. Unlike the Ohmic contacts, the gate metallisation is not annealed; preserving the Schottky barrier height is essential for minimising leakage current and maintaining effective depletion control over the channel.

After metallisation, the resist was removed using solvent lift-off to reveal the final device structure. A small number of lift-off defects (primarily isolated metallic particles) were observed under the microscope and subsequently removed using brief, gentle ultrasonic cleaning. The removal of these artefacts is essential for device reliability: they can otherwise cause electrical short circuits, particularly around the sensitive gate edges.

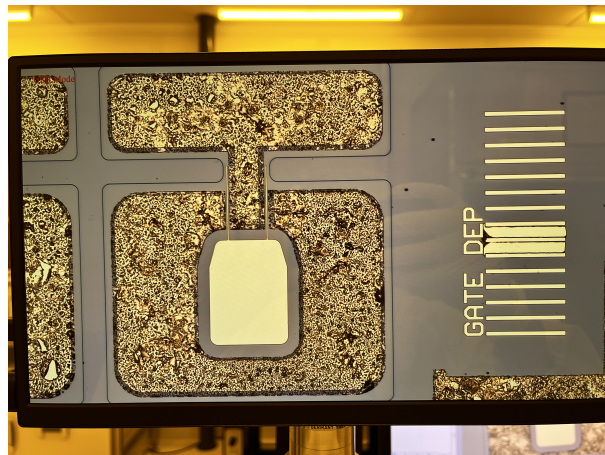


Figure 3.5: Gate contacts after lift-off.

The completed devices were examined using optical microscopy to ensure process integrity. The mesas exhibited clean, well-defined sidewalls; the Ohmic pads were properly positioned; and the gate electrodes showed sharp, uninterrupted edges with no evidence of bridging or residual flakes. Overall, the fabrication process yielded a high-quality array of depletion-mode HEMTs

suitable for subsequent electrical characterisation.

CHAPTER 4

Electrical Characterisation and Critical Analysis

4.1 Measurement Setup

4.2 Output Characteristics (ID–VD)

4.3 Transfer Characteristics (ID–VG)

4.4 Gate Leakage Behaviour

4.5 Threshold-Voltage Extraction

4.6 Variability, Non-Idealities and Interpretation

4.7 Implications for Process Control and Device Behaviour

CHAPTER 5

Exploration of Enhancement-Mode HEMT Devices

- 5.1 Motivation for Enhancement-Mode Operation
- 5.2 Mechanisms for Achieving Enhancement Mode
- 5.3 Challenges of Gate Recessing Without an Etch-Stop Layer
- 5.4 Proposed Fabrication Strategy for Semester Two
- 5.5 Ideal Outcomes and Target Behaviour
- 5.6 Logic Motivation and Stretch Goal

CHAPTER 6

Reflection and Project Management

6.1 Summary of Progress

6.2 Challenges and Adaptations

6.3 Skills Developed

6.4 Lessons for Semester Two

CHAPTER 7

Semester Two Plan

7.1 Gantt Chart

7.2 Planned Technical Deliverables

7.3 Risk Mitigation Strategy

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APPENDIX A

Additional Figures

APPENDIX B

Code and Process Flow